

Structured Pruning and Lightweight Adaptation of Foundation Models for Resource-Efficient Earth Observation

Aniket Ramchandra Munde¹, Kartheek Kumar Reddy Nareddy², Julia Niebling²

¹Hof University of Applied Sciences, Hof, Germany

²Institute of Data Science, German Aerospace Center (DLR), Jena, Germany

aniket.munde@hof-university.de, {kartheek.nareddy, julia.niebling}@dlr.de

Abstract

Structured pruning has gained attention as a promising solution for the cost-effective deployment of large language, vision-language, and domain-specific foundation models. However, most prior work has focused on general-purpose language models and often lacks a systematic comparison of pruning strategies across model scales, architectures, and real-world multimodal applications. In this paper, we conduct a comprehensive evaluation of structured pruning methods, focusing on depth pruning, width pruning, and post-pruning recovery with parameter-efficient fine-tuning. We evaluate multiple open-source models, including Qwen3-0.6B, Qwen3-4B-Instruct, Qwen3-8B, LLaVA-OneVision-0.5B, and Phi-4-Mini-Instruct, across intrinsic metrics and downstream benchmarks such as BBH, MATH Hard, MuSR, and MMLU-Pro. Our findings reveal that (1) depth pruning generally provides a more stable trade-off between efficiency and performance than width pruning; (2) width pruning can reduce memory usage, but often gives limited throughput gains and less predictable accuracy retention; (3) reasoning-intensive benchmarks, particularly mathematical reasoning tasks, are more sensitive to structured pruning than some other evaluation tasks; (4) LoRA-based post-pruning adaptation can substantially recover performance after moderate pruning; and (5) pruning behaviors observed in general-purpose models transfer to Earth observation foundation models, as demonstrated on TerraMind, where depth pruning reduces latency and GPU memory usage while maintaining competitive EuroSAT and Bigearthnet task accuracy.

1 Introduction

Despite the strong performance of recent large language models (LLMs) and foundation models on language understanding, reasoning, and multimodal tasks (Zhu et al., 2023; Vaswani et al., 2017),

their deployment under resource constraints is still difficult. These models require substantial memory and computation, which limits their use in edge devices, mobile systems, and applications with strict latency requirements. The issue is particularly relevant in scientific domains such as Earth observation, where large-scale geospatial data has to be processed under practical limits on memory, runtime, and energy consumption (Lu and et al., 2024; Liang et al., 2023).

Model compression is one of the main directions for making foundation models more usable in such settings. Pruning, quantization, and knowledge distillation have all been studied as ways to reduce model size and inference cost (Muralidharan et al., 2024; Frantar et al., 2022; Xiao et al., 2023). Structured pruning removes complete parts of the architecture, such as Transformer layers, attention heads, or feed-forward channels and results in smaller dense models that can still be executed using standard hardware acceleration libraries (Ma et al., 2023; Ling et al., 2024). Structured pruning is more likely to produce measurable gains in latency and memory usage without relying on specialized sparse kernels (Xu et al., 2025; Kim et al., 2024). Depth pruning removes whole Transformer layers and takes advantage of the fact that not all layers contribute equally to the final model behavior (Lu and et al., 2024; Siddiqui and et al., 2024). Width pruning, by comparison, removes channels or internal dimensions, and is harder to apply cleanly because of the dependencies between attention, projection, and feed-forward components (Xu et al., 2025; Ma et al., 2023; Gao et al., 2024).

In this paper, we present a systematic evaluation of structured pruning and post-pruning adaptation for language, vision-language, and Earth observation foundation models. In particular, we aim to address the following research questions (RQs): **RQ1:** How do depth pruning and width pruning differ in terms of performance retention

across language-only and multimodal foundation models? **RQ2:** How does structured pruning affect downstream reasoning performance on benchmarks such as BBH, MATH Hard, MuSR, and MMLU-Pro? **RQ3:** How do model scale and architecture influence pruning robustness across Qwen3, Phi-4-Mini-Instruct, and LLaVA-OneVision models? **RQ4:** To what extent can parameter-efficient fine-tuning, particularly LoRA, recover performance after structured pruning? **RQ5:** Do pruning behaviors observed in general-purpose language and vision-language models transfer to domain-specific Earth observation foundation models such as TerraMind, and can pruning improve latency and memory efficiency while maintaining competitive EuroSAT accuracy?

We evaluate several open-source models, including Qwen3 (Yang et al., 2025) variants at different scales, LLaVA-OneVision-0.5B (Li et al., 2024), and Phi-4-Mini-Instruct (Abdin et al., 2024). The evaluation combines prompt-based forced continuation (Wee et al., 2025), dataset-based analysis on WikiText-2 (Merity et al., 2016), and downstream benchmark evaluation on BBH, MATH Hard, MuSR, and MMLU-Pro. We further measure inference latency, time-to-first-token, throughput, and peak GPU memory usage. Beyond general-purpose models, we extend the study to TerraMind, a multimodal foundation model for Earth observation (Jakubik et al., 2025). TerraMind processes heterogeneous geospatial inputs such as multispectral optical imagery and radar data, making it a representative case study for real-world scientific deployment.

Our key findings are as follows:

- Depth pruning generally provides a more stable trade-off between efficiency and performance than width pruning across language and multimodal models.
- Width pruning can reduce memory usage in larger models, but it often provides limited throughput improvement and less predictable accuracy retention due to architectural dependencies.
- Reasoning-intensive tasks, especially mathematical reasoning, are more sensitive to structured pruning than some other benchmarks.
- LoRA-based post-pruning adaptation can substantially recover performance after moderate

pruning, although recovery becomes limited under aggressive compression (Hu et al., 2021; Dettmers et al., 2023).

- The Earth observation usecase shows that pruning behavior observed in general-purpose models can transfer to Earth observation foundation models, where depth pruning reduces latency and GPU memory usage while maintaining competitive EuroSAT and Bigearthnet accuracy.

2 Related Work

Model compression has become an active direction for reducing the cost of large language models and foundation models. Common approaches include pruning, quantization, and knowledge distillation (Muralidharan et al., 2024). Quantization methods such as GPTQ and SmoothQuant reduce numerical precision during inference (Frantar et al., 2022; Xiao et al., 2023), while adaptation and recovery methods such as LoRA and QLoRA reduce the cost of fine-tuning large models (Hu et al., 2021; Dettmers et al., 2023). LLM-Pruner estimates the importance of coupled structures and prunes them using dependency-aware grouping (Ma et al., 2023). SlimGPT uses second-order information inspired by Optimal Brain Surgeon for structured pruning with reduced accuracy loss (Ling et al., 2024). DISP-LLM relaxes the fixed-width assumption across Transformer blocks and allows dimension-independent pruning, although the resulting architecture becomes less straightforward to handle (Gao et al., 2024). Other work, such as 2SSP, considers staged pruning of feed-forward and attention modules to limit error accumulation during compression (Sandri et al., 2025a).

Evaluation and Earth observation models:

Most pruning studies focus on general-purpose language models and usually evaluate a limited number of architectures or tasks. This makes it difficult to compare depth and width pruning under the same experimental setting, especially across different model sizes and model families. Post-pruning recovery is also not always studied in detail, although lightweight adaptation can be important for restoring part of the lost performance after compression (Hu et al., 2021; Dettmers et al., 2023). Our work follows this direction by evaluating structured pruning across Qwen3, Phi-4-Mini-Instruct, and LLaVA-OneVision models, while also considering downstream reasoning performance and practical

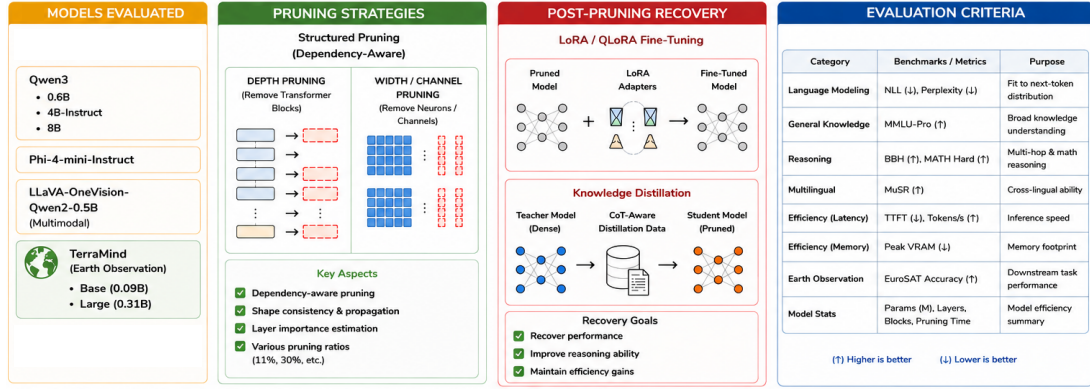


Figure 1: Methodology overview of the proposed approach. The figure summarizes the evaluated model families, structured depth and width pruning strategies, post-pruning recovery using Parameter Efficient Fine-Tuning and knowledge distillation, and metrics used for evaluation.¹

efficiency.

In Earth observation, foundation models are increasingly used to process heterogeneous satellite data. TerraMind is designed for multi-sensor and multimodal geospatial inputs, including Sentinel-1 radar and Sentinel-2 multispectral imagery (Jakubik et al., 2025). Related systems such as FAST-EO study multimodal alignment between satellite imagery and text, supporting tasks such as captioning and environmental monitoring (Sandri et al., 2025b). These models are useful for EO applications, but they are still computationally demanding and less studied from the point of view of compression.

3 Methodology

This work studies structured pruning for improving the efficiency of language, vision-language, and Earth observation foundation models. The focus is on two forms of structured pruning: depth pruning and width pruning. We also consider dependency-aware pruning rules, since many Transformer components are coupled and cannot be removed independently. The summary of the methodology can be found in Fig. 1.

3.1 Structured Pruning

Structured pruning removes complete model components, such as Transformer layers, attention heads, projection channels, or feed-forward dimensions. This produces smaller dense models that can still run on standard hardware accelerators

(Ma et al., 2023; Ling et al., 2024). This is different from unstructured pruning, where individual weights are removed and the resulting sparsity may not directly translate into practical speedups without specialized kernels (Xu et al., 2025).

3.2 Depth Pruning

Depth pruning reduces model depth by removing complete Transformer blocks (Lu et al., 2023). Given a model with L layers, we estimate the importance of each layer using an ablation-based sensitivity test. Each layer is temporarily removed, and the resulting change in negative log-likelihood (NLL) and perplexity (PPL) is measured (Gehrmann et al., 2024; Lu and et al., 2024; Siddiqui and et al., 2024). Layers that lead to only small degradation are treated as better pruning candidates and relying solely on local metrics such as PPL can be misleading because layers showing limited isolated impact may still participate in important architectural interactions.

3.3 Width Pruning

Width pruning reduces the internal dimensionality of Transformer layers. It targets structures such as attention heads, projection channels, and feed-forward neurons (Xu et al., 2025). In our setting, pruning candidates are identified using magnitude-based criteria, where smaller-magnitude parameters are assumed to contribute less to model behaviour. Attention projections, feed-forward blocks, residual paths, and normalization layers are often linked through shared dimensions. If these dependencies are ignored, pruning can introduce invalid tensor shapes or unstable model behaviour (Gao et al., 2024).

¹This figure was generated with the assistance of generative AI tools and was subsequently reviewed, edited, and finalized by the authors.

3.4 Dependency-Aware Pruning

To preserve architectural validity, we apply dependency-aware pruning rules. Instead of pruning each component separately, linked structures are pruned in a coordinated way. For example, attention heads and their corresponding projection dimensions must be removed consistently across the query, key, value, and output projections (Ma et al., 2023). Similar constraints are needed for feed-forward layers, where input and output dimensions are coupled across paired projections. We also use simple safety constraints, such as minimum channel thresholds, to avoid excessive removal of representational capacity from a layer.

3.5 Post-Pruning Recovery

Pruning can reduce model quality, especially at higher pruning ratios. To recover part of the lost performance, we use post-pruning recovery methods based on parameter-efficient fine-tuning and knowledge distillation (Yu et al., 2023; Hu et al., 2021; Dettmers et al., 2023). LoRA updates only a small number of additional trainable parameters while keeping most of the original model fixed. This makes it suitable for lightweight recovery after pruning. Knowledge distillation can further guide the pruned model by transferring useful behaviour from the original dense model.

3.6 Extension to Multimodal Foundation Models

The same pruning ideas are extended to TerraMind, a multimodal foundation model for Earth observation. Compared with text-only models, TerraMind has additional dependencies between modality-specific and shared representations, since it processes optical and radar inputs within a common architecture. Thus, depth pruning is the optimal choice in the Earth observation usecase. Removing complete Transformer blocks keeps the remaining internal dimensions unchanged and reduces the risk of disrupting modality alignment. This allows us to test whether pruning behaviour observed in language models also appears in a domain-specific Earth observation model.

4 Experimental Setup

4.1 Models

We evaluate structured pruning on several open-source language and multimodal models. The language model experiments include Qwen3-0.6B,

Qwen3-4B-Instruct, Qwen3-8B, and Phi-4-Mini-Instruct. LLaVA-OneVision-0.5B is included as a compact vision-language model. These models cover different parameter scales and model families, allowing us to compare pruning behaviour across architectures. For the Earth observation case study, we use TerraMind, a domain-specific multimodal foundation model (Jakubik et al., 2025). TerraMind is used to examine whether pruning trends observed in general-purpose models also transfer to geospatial models.

4.2 Benchmarks and Metrics

Model quality is evaluated using both intrinsic and downstream metrics. For intrinsic evaluation, we report negative log-likelihood (NLL) and perplexity (PPL). NLL is used as the main metric for pruning-sensitive comparisons, while PPL is retained for reporting and comparison with standard language-modeling practice. Downstream reasoning performance is measured using lm-evaluation-harness (EleutherAI, 2024). The benchmark set includes BBH (Suzgun et al., 2022), MMLU-Pro (Wang et al., 2024), MuSR (Sprague et al., 2023), and Math Hard. These tasks cover logical reasoning, knowledge, multi-step problem solving, and mathematical reasoning. We also report practical efficiency metrics, including time-to-first-token (TTFT), generation throughput, and peak GPU memory usage.

4.3 Prompt-Based and Dataset-Based Evaluation

Two evaluation modes are used. First, prompt-based forced-continuation evaluation is applied for controlled pruning experiments (Wee et al., 2025). Unless stated otherwise, all models are evaluated with the same 0-shot probe prompt:

“Describe the city of Stuttgart in a few sentences.”

The continuation length is fixed to 300 tokens, and the random seed is set to 123. Second, dataset-based evaluation is performed on WikiText-2 (Merity et al., 2016). We use a sliding-window setup with sequence length 1024 and stride values of 1024 or 512. This setting is used for NLL and PPL measurements and for layer sensitivity analysis during depth pruning.

4.4 Pruning and Recovery Protocol

Depth and width pruning are applied according to the procedure described in Section 3. For each model, we first evaluate the dense baseline, then apply structured pruning, and finally evaluate the pruned model using the same benchmark and efficiency pipeline. For post-pruning recovery, LoRA modules are inserted into selected attention and feed-forward projections (Hu et al., 2021; Dettmers et al., 2023). Most original model weights remain fixed during this stage. This setup allows us to measure whether lightweight adaptation can recover performance without full retraining.

4.5 Earth Observation

TerraMind is evaluated on EuroSAT for land-cover classification. The model expects Sentinel-2 Level-2A multispectral inputs and Sentinel-1 radar inputs. Since EuroSAT does not contain radar measurements, a zero-filled Sentinel-1 tensor is used to preserve the multimodal input interface. RGB images are converted to a 12-channel Sentinel-2-like representation by channel repetition. Inputs with 13 channels are reduced to 12 channels for consistency, and all images are resized to 224×224 pixels. In the baseline setting, the TerraMind backbone is frozen and used as a feature extractor. A linear classification head is trained on top of the extracted embeddings. For the pruned setting, structured depth pruning is applied to the backbone, and the same linear probing protocol is used. For recovery, LoRA modules and the classification head are trained while the original backbone weights remain fixed. The LoRA adapters use rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.05, and are inserted into attention and MLP projection layers. The recovery stage trains only the LoRA parameters and classification head using bfloat16 precision, with learning rates of 5×10^{-4} for LoRA and 10^{-3} for the head.

4.6 Reproducibility Details

All language-model experiments are run on the Kratos HPC cluster using SLURM. Standard pruning and evaluation jobs use a single Ampere GPU with 100 GB RAM. Larger benchmark runs use vLLM with lm-evaluation-harness on multiple Ampere GPUs (Kwon et al., 2023). The main software environment uses Python 3.10.14 with isolated dependencies. TerraMind experiments use a separate Python 3.11 environment for compatibility

with the required geospatial and model dependencies. The same workflow is followed across experiments: baseline evaluation, structured pruning, and post-pruning evaluation with optional recovery. Fixed prompts, controlled seeds, and identical preprocessing are used where possible, so that performance differences can be attributed mainly to pruning and recovery choices.

5 Results and Analysis

We present the experimental results of structured pruning across multiple language and multimodal models. The analysis focuses on the trade-off between efficiency and downstream performance, while comparing the behavior of depth and width pruning under a unified evaluation framework. The summary of the results can be found in Fig. 2.

5.1 LLaVA-OneVision

LLaVA-OneVision-0.5B (Li et al., 2024) is used to test pruning in a compact vision-language model. Table 1 shows that depth pruning keeps the BBH score unchanged and gives a small gain on MuSR. Width pruning remains close to the baseline, but shows a slight drop on BBH. This suggests that some redundancy exists even in the compact multimodal setting, while channel-level pruning is still a little less stable.

Table 1: Benchmark performance of LLaVA-OneVision-0.5B under depth and width pruning.

Configuration	BBH $\uparrow$	MATH Hard $\uparrow$	MuSR $\uparrow$
Baseline	28.78	0.00	36.64
Depth-pruned (22/24)	28.78	0.00	37.57
Width-pruned (2%)	28.21	0.00	37.30

5.2 Qwen3

The Qwen3 family (Yang et al., 2025) is evaluated at three scales: 0.6B, 8B, and 4B-Instruct. The results show that pruning behaviour changes with scale and tuning. Smaller models show modest changes, while the instruction-tuned and larger variants are more sensitive, especially on mathematical reasoning. For Qwen3-0.6B, both pruning strategies keep BBH close to the baseline, as shown in Table 2. MuSR improves after pruning, with the best score obtained by width pruning. MATH Hard, however, drops under both pruning settings, indicating that even mild structural changes can affect mathematical reasoning.

Table 3: Benchmark performance of Qwen3-8B under pruning and LoRA recovery.

Configuration	BBH $\uparrow$	MATH Hard $\uparrow$	MuSR $\uparrow$	MMLU-Pro $\uparrow$
Baseline	29.30	17.30	40.60	11.36
Depth-pruned (11%)	31.05	3.10	38.36	12.36
Depth-pruned (30%)	28.09	0.08	36.11	11.70
Width-pruned (11.1%)	25.93	1.81	41.67	N/A
Depth-pruned (11%) + LoRA	43.59	7.80	32.54	N/A
Depth-pruned (30%) + LoRA	36.16	1.59	35.71	N/A

Table 4: Benchmark performance of Qwen3-4B-Instruct under different depth-pruning choices.

Configuration	BBH $\uparrow$	MATH Hard $\uparrow$	MuSR $\uparrow$	MMLU-Pro $\uparrow$
Baseline	54.54	45.85	40.61	43.94
Depth-pruned (30%)	29.25	1.44	36.24	12.16
WT2-guided depth-pruned (30%)	29.37	1.28	38.49	N/A
WT2-random depth-pruned (30%)	30.29	0.98	38.49	N/A

Table 2: Benchmark performance of Qwen3-0.6B across baseline and pruned variants.

Configuration	BBH $\uparrow$	MATH Hard $\uparrow$	MuSR $\uparrow$
Baseline	28.33	8.16	31.35
Depth-pruned (~5%)	27.76	3.55	33.20
Width-pruned (2%)	28.36	4.91	36.38

Qwen3-8B shows clearer task-level differences. As Table 3 shows, 11% depth pruning keeps BBH competitive and slightly improves MMLU-Pro, but MATH Hard drops sharply. With 30% depth pruning, the degradation becomes broader. Width pruning gives the best MuSR score, but performs worse on BBH and MATH Hard. LoRA recovery improves BBH strongly after pruning, although recovery on MATH Hard and MuSR remains limited.

For Qwen3-4B-Instruct, pruning is more damaging. Table 4 shows large drops on BBH, MATH Hard, and MMLU-Pro under 30% depth pruning. MuSR is less affected. WT2-guided and random depth pruning give similar scores, suggesting that, at this pruning level, the pruning ratio matters more than the exact layer selection rule.

5.3 Phi-4-Mini-Instruct

Phi-4-Mini-Instruct (Abdin et al., 2024) provides another compact instruction-tuned case. Table 5 shows that 11% depth pruning leads to clear degradation on BBH, MATH Hard, and MMLU-Pro. MuSR is more stable and slightly improves. The pattern is close to what is observed in Qwen3 models: mathematical and broad reasoning scores are more fragile, while MuSR is less affected.

Table 5: Benchmark performance of Phi-4-Mini-Instruct before and after depth pruning.

Configuration	BBH $\uparrow$	MATH Hard $\uparrow$	MuSR $\uparrow$	MMLU-Pro $\uparrow$
Baseline	56.40	16.80	39.20	39.30
Depth-pruned (11%)	32.32	0.00	40.48	22.83

RQ1: How do depth and width pruning differ across models?

Depth pruning is more stable in most evaluated settings. In LLaVA-OneVision-0.5B (Li et al., 2024), removing two Transformer layers keeps BBH unchanged at 28.78 and slightly improves MuSR from 36.64 to 37.57. Width pruning gives a close MuSR score, but the BBH score drops from 28.78 to 28.21. A similar tendency appears for Qwen3-8B (Yang et al., 2025), where moderate depth pruning gives 31.05 on BBH, while width pruning gives 25.93.

Depth pruning removes full blocks and keeps the remaining hidden dimensions unchanged, which makes the pruned model easier to execute. Width pruning can be useful for reducing memory, but it modifies internal dimensions and depends more strongly on structural coupling between attention projections, feed-forward layers, and residual paths. Depth pruning gives a more dependable accuracy–efficiency trade-off across the evaluated language and multimodal models. Width pruning is more useful when GPU memory is the main constraint.

RQ2: How does pruning affect downstream reasoning performance?

MATH Hard is the most sensitive task to pruning in our experiments. For Qwen3-8B, 11% depth pruning reduces MATH Hard from 17.30 to 3.10, even though BBH remains close to the baseline and MMLU-Pro slightly improves. Phi-4-Mini-Instruct (Abdin et al., 2024) shows a similar pattern: MATH Hard drops from 16.80 to 0.00 after depth pruning, while MuSR slightly increases from 39.20 to 40.48.

MuSR is less fragile in several settings. Some pruned models remain close to the dense baseline, and a few even improve slightly. This indicates that pruning does not merely lower all reasoning scores uniformly. It affects tasks depending on how strongly they rely on precise intermediate computation. Mathematical reasoning is more vulnerable to structured pruning than BBH or MuSR. Moderate pruning can preserve parts of downstream reasoning ability, but MATH Hard shows clear sensitivity.

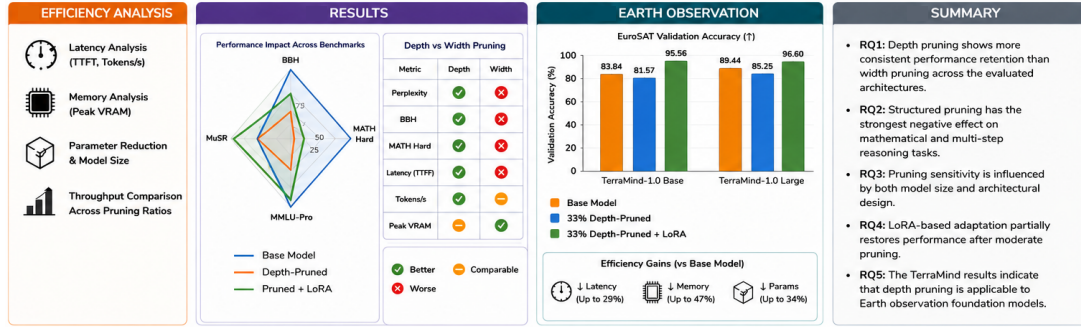


Figure 2: Results overview across language, vision-language, and Earth observation foundation models. The figure summarizes reasoning accuracy, language modeling quality, EuroSAT performance, latency, memory usage, model statistics, and key findings.

RQ3: How do model scale and architecture influence pruning robustness?

The results vary with both scale and architecture. Qwen3-0.6B shows only modest changes on BBH and MuSR, although MATH Hard still drops after pruning. Qwen3-4B-Instruct is much more sensitive under 30% depth pruning, with large declines on BBH, MATH Hard, and MMLU-Pro. Phi-4-Mini-Instruct also loses performance on BBH and MMLU-Pro after depth pruning, despite a small gain on MuSR.

Parameter count alone does not fully explain pruning robustness. Instruction tuning, layer redundancy, and model-specific design choices appear to matter. For Qwen3-4B-Instruct, WT2-guided and random depth pruning produce similar results at 30% pruning. At this level, the amount of removed capacity seems to dominate the exact layer selection rule. Model size is important, but architecture and pruning ratio also shape the final accuracy. Instruction-tuned models can degrade sharply when pruning becomes aggressive.

RQ4: Can LoRA recover performance after pruning?

LoRA recovery improves some pruned models, but it does not recover all tasks equally. For Qwen3-8B, LoRA after 11% depth pruning improves BBH from 31.05 to 43.59 and raises MATH Hard from 3.10 to 7.80. The same adaptation reduces MuSR from 38.36 to 32.54. With 30% depth pruning, LoRA still improves BBH from 28.09 to 36.16, while MATH Hard remains low. These results suggest that lightweight adaptation can restore useful

capabilities after moderate pruning. It is less effective when the pruning ratio is high or when the task requires precise multi-step reasoning. LoRA can recover a substantial part of BBH performance after pruning, but the recovery is task-dependent. Mathematical reasoning remains difficult to restore after structural compression.

5.4 RQ5: Do pruning trends transfer to Earth observation foundation models?

Pruning and finetuning experiments with TerraMind (Jakubik et al., 2025) show that structured depth pruning also transfers to a domain-specific multimodal model. For TerraMind-1.0-Base, pruning reduces the number of blocks from 12 to 8 and parameters from 87.71M to 59.39M. EuroSAT accuracy decreases from 83.84% to 81.57% and BigEarthNet accuracy from 92.94% to 91.67%, while latency improves from 4.59 ms to 3.14 ms and GPU memory decreases from 312.5 MB to 196.0 MB. For TerraMind-1.0-Large, pruning reduces latency from 9.13 ms to 7.66 ms and GPU memory from 1119.1 MB to 642.1 MB, with EuroSAT accuracy decreasing from 89.44% to 85.25% and BigEarthNet accuracy only slightly decreasing from 93.24% to 93.03%. LoRA fine-tuning recovers most of the accuracy loss and raises validation accuracy beyond the frozen dense baseline. On EuroSAT, pruned LoRA fine-tuning reaches 95.56% for TerraMind-1.0-Base and 96.60% for TerraMind-1.0-Large. On BigEarthNet, pruned LoRA fine-tuning reaches 93.56% and 95.04%, respectively, remaining close to the dense LoRA results of 94.99% and 95.15%.

5.5 Efficiency Analysis

Beyond model quality, we evaluate the efficiency impact of structured pruning using system-level

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Table 6: TerraMind validation accuracy and efficiency before and after depth pruning. EuroSAT and BigEarthNet report downstream validation accuracy. LoRA fine-tuning rows include the adapted model parameters and measured inference efficiency after recovery.

Model	Configuration	EuroSAT Acc. (%) ↑	BigEarthNet Acc. (%) ↑	Blocks	Params (M)	Latency (ms) ↓	GPU Mem. (MB) ↓
TerraMind-1.0-Base	Baseline	83.84	92.94	12	87.71	4.59	312.5
TerraMind-1.0-Base	Depth-pruned	81.57	91.67	8	59.39	3.14	196.0
TerraMind-1.0-Base	Dense + LoRA FT	97.23	94.99	12	89.00	4.72	324.2
TerraMind-1.0-Base	Pruned + LoRA FT	95.56	93.56	8	60.25	3.21	201.4
TerraMind-1.0-Large	Baseline	89.44	93.24	24	305.66	9.13	1119.1
TerraMind-1.0-Large	Depth-pruned	85.25	93.03	20	255.33	7.66	642.1
TerraMind-1.0-Large	Dense + LoRA FT	98.30	95.15	24	309.05	9.37	1135.7
TerraMind-1.0-Large	Pruned + LoRA FT	96.60	95.04	20	257.59	7.83	648.6

Table 7: Relative efficiency changes after pruning across different models and pruning strategies. Depth pruning consistently reduces TTFT and improves token generation throughput, whereas width pruning primarily reduces peak VRAM usage with limited throughput gains.

Model	Method	TTFT Change (%) ↓	Tok/s Change (%) ↑	Peak VRAM Reduction (%) ↑
LLaVA-OneVision-0.5B	Depth	-10.71	+5.66	0.00
LLaVA-OneVision-0.5B	Width	+12.50	0.00	0.00
Qwen3-0.6B	Depth	-5.88	+9.75	0.00
Qwen3-0.6B	Width	0.00	0.00	-0.88
Qwen3-4B-Instruct	Depth 30%	-22.50	+40.86	-0.13
Qwen3-4B-Instruct	Width 30%	+4.76	+0.40	+20.18
Qwen3-8B	Depth 30%	-19.51	+41.09	-0.13
Qwen3-8B	Width 30%	+4.76	+1.19	+19.63
Phi-4-mini-Instruct	Depth 11%	-3.57	+12.60	-0.42

inference metrics, including time-to-first-token (TTFT), generation throughput, and peak GPU memory usage (Sheng et al., 2023). Table 7 reports the relative changes after pruning. Depth pruning consistently improves inference speed. For Qwen3-4B-Instruct, 30% depth pruning reduces TTFT by 22.50% and improves throughput by 40.86%. A similar trend appears for Qwen3-8B, where 30% depth pruning reduces TTFT by 19.51% and improves throughput by 41.09%. These results show that removing full Transformer layers translates directly into faster generation.

Width pruning provides limited throughput gains but reduces peak VRAM more effectively in larger models. For Qwen3-4B-Instruct, 30% width pruning reduces peak VRAM by 20.18%, and for Qwen3-8B it reduces peak VRAM by 19.63%. However, the corresponding throughput improvements are only 0.40% and 1.19%, respectively.

6 Conclusion

We studied structured pruning for improving the efficiency of language, vision-language, and Earth observation foundation models. Experiments across Qwen3, LLaVA-OneVision, Phi-4-Mini-

Instruct, and TerraMind show that depth pruning is generally more stable than width pruning, since removing complete Transformer blocks gives clearer latency and throughput gains without changing the internal dimensions of the remaining model. Width pruning can reduce memory usage, but it is more difficult to control because channel-level changes interact with attention projections, feed-forward layers, and other structural dependencies. The results also show that pruning affects tasks differently: mathematical and structured reasoning benchmarks are more sensitive to compression, while some other reasoning tasks remain comparatively stable under moderate pruning. LoRA-based recovery improves post-pruning performance in several cases, although the recovery is limited when compression becomes aggressive. The Earth observation usecase suggests that these trends extend beyond text-only models to domain-specific multimodal foundation models, where depth pruning reduces latency and memory usage while preserving useful representations for Earth observation tasks. Future work can extend this analysis to richer multimodal datasets and more detailed studies of cross-modal pruning dependencies.

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